

Decision and Risk Analysis

Lecture 6: **Conjugate Families in Bayesian Modeling**

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What is the typical number of scam/fraud calls received per day?

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Let λ denote the daily rate of calls.

Before collecting data, suppose we believe:

- the rate is probably around 5 calls/day;
- values between 2 and 7 calls/day are plausible;
- the rate must be positive: $\lambda > 0$.

Then we observe four days of data:

$$\vec{y} = (6, 2, 2, 1), \quad n = 4, \quad \sum y_i = 11, \quad \bar{y} = 2.75.$$

Combine prior belief with data to estimate a posterior model for λ .

Bayes' Rule: The Core Update

Bayesian inference combines two ingredients:

$$\text{posterior} \propto \text{prior} \times \text{likelihood}.$$

For a parameter θ and observed data y :

$$f(\theta \mid y) \propto f(\theta)L(\theta \mid y).$$

Conjugacy: A prior is conjugate for a likelihood when the posterior belongs to the same model family as the prior.

Why? Conjugate priors give posterior models that are easy to calculate and easy to interpret.

Beta-Binomial Conjugacy

When Y is a number of successes in n trials:

$$Y \mid \pi \sim \text{Bin}(n, \pi), \quad \pi \sim \text{Beta}(\alpha, \beta).$$

The posterior is still Beta:

$$\pi \mid y \sim \text{Beta}(\alpha + y, \beta + n - y).$$

Interpretation:

- α and β act like prior successes and failures.
- y and $n - y$ are observed successes and failures.
- Large prior hyperparameters make the prior harder to move.

Why Non-Conjugate Priors Can Be Awkward

Suppose $Y \mid \pi \sim \text{Bin}(50, \pi)$ and we observe $y = 10$.
A non-Beta prior such as

$$f(\pi) = e - e^\pi, \quad 0 \leq \pi \leq 1$$

produces

$$f(\pi \mid y = 10) \propto (e - e^\pi) \pi^{10} (1 - \pi)^{40}.$$

This is a valid posterior kernel, but it is not a familiar distribution.
To normalize it, we would need

$$\int_0^1 (e - e^\pi) \pi^{10} (1 - \pi)^{40} d\pi.$$

Conjugacy avoids this type of integral in many common models.

Poisson Data Model for Counts

For the scam-call example, the data are counts with no fixed upper bound. A natural model is

$$Y_i \mid \lambda \stackrel{\text{ind}}{\sim} \text{Pois}(\lambda), \quad i = 1, \dots, n.$$

The Poisson probability mass function is

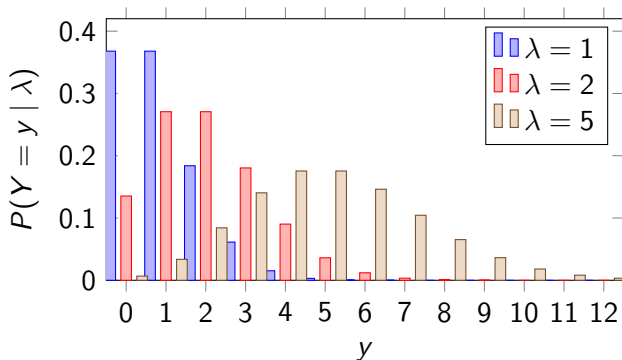
$$f(y_i \mid \lambda) = \frac{\lambda^{y_i} e^{-\lambda}}{y_i!}, \quad y_i = 0, 1, 2, \dots$$

For a Poisson variable:

$$E(Y_i \mid \lambda) = \text{Var}(Y_i \mid \lambda) = \lambda.$$

Poisson Models with Different Rates

Poisson pmfs: larger λ shifts mass right and increases spread



Deriving the Poisson Likelihood

For independent counts $Y_1, \dots, Y_n$:

$$f(\vec{y} \mid \lambda) = \prod_{i=1}^n \frac{\lambda^{y_i} e^{-\lambda}}{y_i!}.$$

Simplify the product:

$$\begin{aligned} f(\vec{y} \mid \lambda) &= \frac{\lambda^{y_1} \lambda^{y_2} \dots \lambda^{y_n} e^{-\lambda} e^{-\lambda} \dots e^{-\lambda}}{y_1! y_2! \dots y_n!} \\ &= \frac{\lambda^{\sum y_i} e^{-n\lambda}}{\prod_{i=1}^n y_i!}. \end{aligned}$$

As a function of λ :

$$L(\lambda \mid \vec{y}) \propto \lambda^{\sum y_i} e^{-n\lambda}.$$

Likelihood for the Scam-Call Data

Observed data:

$$\vec{y} = (6, 2, 2, 1), \quad n = 4, \quad \sum y_i = 11.$$

Therefore:

$$L(\lambda \mid \vec{y}) = \frac{\lambda^{11} e^{-4\lambda}}{6!2!2!1!} \propto \lambda^{11} e^{-4\lambda}.$$

The likelihood is maximized near

$$\hat{\lambda} = \bar{y} = \frac{11}{4} = 2.75.$$

Interpretation: Rates near 2.75 calls/day are most compatible with the observed data.

Gamma Prior for a Positive Rate

A rate parameter must satisfy $\lambda > 0$. The Gamma distribution is a natural prior:

$$\lambda \sim \text{Gamma}(s, r), \quad s > 0, \quad r > 0.$$

Using the shape-rate parameterization:

$$f(\lambda) = \frac{r^s}{\Gamma(s)} \lambda^{s-1} e^{-r\lambda}, \quad \lambda > 0.$$

Important summaries:

$$E(\lambda) = \frac{s}{r}, \quad \text{Mode}(\lambda) = \frac{s-1}{r}, \quad \text{Var}(\lambda) = \frac{s}{r^2}.$$

Tuning the Prior in the Example

Prior belief: λ is around 5, with plausible values roughly between 2 and 7.

Choose

$$\lambda \sim \text{Gamma}(10, 2).$$

Then

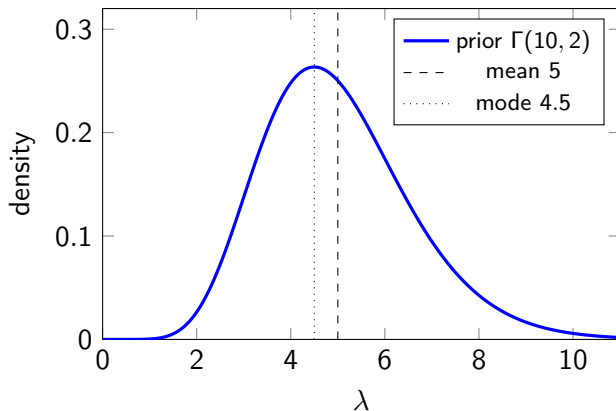
$$E(\lambda) = \frac{10}{2} = 5, \quad \text{Mode}(\lambda) = \frac{10 - 1}{2} = 4.5,$$

$$\text{SD}(\lambda) = \sqrt{\frac{10}{2^2}} = 1.581.$$

This prior is centered near 5 but still allows meaningful uncertainty.

Gamma Prior for λ

$\Gamma(10, 2)$ prior: mean 5, mode 4.5



Gamma-Poisson Conjugacy

Model:

$$Y_i \mid \lambda \overset{\text{ind}}{\sim} \text{Pois}(\lambda), \quad \lambda \sim \text{Gamma}(s, r).$$

Prior kernel:

$$f(\lambda) \propto \lambda^{s-1} e^{-r\lambda}.$$

Likelihood kernel:

$$L(\lambda \mid \vec{y}) \propto \lambda^{\sum y_i} e^{-n\lambda}.$$

Posterior kernel:

$$\begin{aligned} f(\lambda \mid \vec{y}) &\propto \lambda^{s-1} e^{-r\lambda} \cdot \lambda^{\sum y_i} e^{-n\lambda} \\ &= \lambda^{s+\sum y_i-1} e^{-(r+n)\lambda}. \end{aligned}$$

Therefore:

$$\lambda \mid \vec{y} \sim \text{Gamma} \left(s + \sum y_i, r + n \right).$$

Posterior Calculation for the Guiding Example

Prior:

$$\lambda \sim \text{Gamma}(10, 2).$$

Data:

$$n = 4, \quad \sum y_i = 11.$$

Update:

$$\lambda \mid \vec{y} \sim \text{Gamma}(10 + 11, 2 + 4) = \text{Gamma}(21, 6).$$

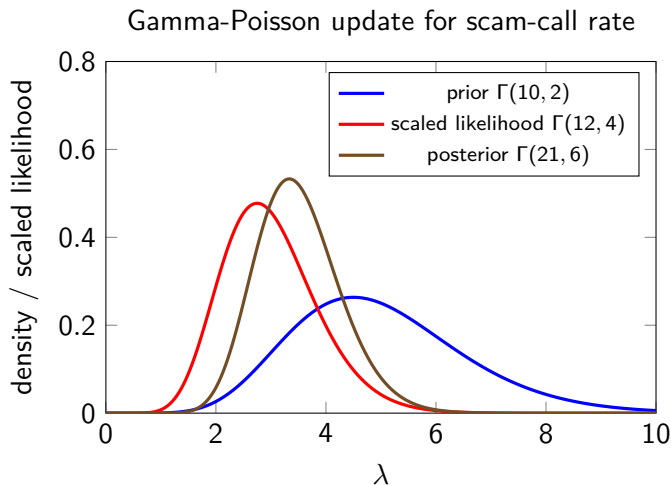
Posterior summaries:

$$E(\lambda \mid \vec{y}) = \frac{21}{6} = 3.5,$$

$$\text{Mode}(\lambda \mid \vec{y}) = \frac{21 - 1}{6} = 3.333,$$

$$\text{SD}(\lambda \mid \vec{y}) = \sqrt{\frac{21}{6^2}} = 0.764.$$

Prior, Scaled Likelihood, and Posterior



Interpreting the Gamma-Poisson Update

Model	Shape	Rate	Mean	SD
Prior	10	2	5.000	1.581
Posterior	21	6	3.500	0.764

The posterior mean is a compromise between:

prior mean = 5 and data mean = 2.75.

The posterior standard deviation is smaller than the prior standard deviation, so the data have reduced uncertainty about λ .

2nd Conjugate Family: Normal-Normal

Now suppose the parameter of interest is a mean μ for continuous measurements.

Example: average hippocampal volume among people with a history of concussions.

Data model with known σ :

$$Y_i \mid \mu \stackrel{\text{ind}}{\sim} \text{N}(\mu, \sigma^2).$$

Prior model:

$$\mu \sim \text{N}(\theta, \tau^2).$$

Because both the likelihood and prior have Normal-shaped kernels in μ , the posterior is also Normal.

The model

We have

$$Y_i \mid \mu \stackrel{\text{ind}}{\sim} N(\mu, \sigma^2), \quad \mu \sim N(\theta, \tau^2).$$

This means:

data Y_i depend on μ ,

and

μ depends on θ .

So this is a two-level model.

Level 1: The observations

Conditional on μ ,

$$Y_i \mid \mu \sim N(\mu, \sigma^2).$$

Interpretation:

$$Y_i = \mu + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2).$$

Each observation varies around the common mean μ .

σ^2 = within-group variation.

Conditional independence

The notation

$$Y_i \mid \mu \stackrel{\text{ind}}{\sim} N(\mu, \sigma^2)$$

means that once we know μ , the observations are independent.

That is,

$$Y_1, Y_2, \dots, Y_n$$

are independent given μ .

But they all share the same unknown mean μ .

Level 2: The mean is random

The mean μ is itself random:

$$\mu \sim N(\theta, \tau^2).$$

Interpretation:

$$\mu = \theta + u, \quad u \sim N(0, \tau^2).$$

So μ varies around the larger population mean θ .

τ^2 = between-group variation.

Putting the levels together

We have

$$Y_i = \mu + \epsilon_i$$

and

$$\mu = \theta + u.$$

Substitute:

$$Y_i = \theta + u + \epsilon_i.$$

So Y_i has two sources of randomness:

$$u \quad \text{and} \quad \epsilon_i.$$

Marginal distribution of Y_i

Since

$$Y_i = \theta + u + \epsilon_i,$$

where

$$u \sim N(0, \tau^2), \quad \epsilon_i \sim N(0, \sigma^2),$$

we get

$$Y_i \sim N(\theta, \tau^2 + \sigma^2).$$

So

$$\text{Var}(Y_i) = \underbrace{\tau^2}_{\text{variation in } \mu} + \underbrace{\sigma^2}_{\text{variation around } \mu}.$$

Are the Y_i 's independent?

Conditional on μ :

Y_i and Y_j are independent.

Marginally:

Y_i and Y_j are not independent.

Why? They share the same random mean μ .

For $i \neq j$,

$$\text{Cov}(Y_i, Y_j) = \tau^2.$$

Correlation between observations

For $i \neq j$,

$$\text{Corr}(Y_i, Y_j) = \frac{\text{Cov}(Y_i, Y_j)}{\sqrt{\text{Var}(Y_i) \text{Var}(Y_j)}}.$$

Since

$$\text{Cov}(Y_i, Y_j) = \tau^2$$

and

$$\text{Var}(Y_i) = \tau^2 + \sigma^2,$$

we get

$$\text{Corr}(Y_i, Y_j) = \frac{\tau^2}{\tau^2 + \sigma^2}.$$

Main idea

The model is

$$Y_i \mid \mu \sim N(\mu, \sigma^2), \quad \mu \sim N(\theta, \tau^2).$$

Where (key points):

σ^2 = variation within a group

τ^2 = variation between groups

θ = overall population mean

This is the basic idea behind hierarchical models and random effects models. We will return to these ideas later in the semester.

Example

Suppose we are studying exam scores.

Y_i = exam score of student i

Students are in the same class.

We assume the class has a true average score:

μ = true class mean

But individual students vary around that class mean.

Level 1: Student scores

We model student scores as

$$Y_i \mid \mu \stackrel{\text{ind}}{\sim} N(\mu, \sigma^2).$$

This means:

$$Y_i = \mu + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2).$$

Here,

σ^2 = variation among students within the class.

The notation

$\stackrel{\text{ind}}{\sim}$

means that the observations are independent, conditional on μ .

So

$$Y_i \mid \mu \stackrel{\text{ind}}{\sim} N(\mu, \sigma^2)$$

means that, once we know the class mean μ , the student scores are independent.

Knowing one student's score does not directly tell us another student's score, after accounting for μ .

One class

Suppose the true class mean is

$$\mu = 80$$

and the within-class standard deviation is

$$\sigma = 5.$$

Then

$$Y_i \mid \mu = 80 \sim N(80, 25).$$

So student scores are centered around 80.

Example scores might be:

$$76, \quad 83, \quad 79, \quad 88, \quad 74.$$

Level 2: Class means vary

Now suppose we are studying many classes.

Different classes may have different true averages.

We model the class mean as

$$\mu \sim N(\theta, \tau^2).$$

Here,

θ = overall average across classes

and

τ^2 = variation between class means.

Numerical example: Many classes

Suppose

$$\theta = 75, \quad \tau = 4.$$

Then

$$\mu \sim N(75, 16).$$

This means class averages tend to be near 75, but not exactly equal.

Some possible class means are:

$$72, \quad 78, \quad 81, \quad 74.$$

Putting the two levels together

The full model is

$$Y_i \mid \mu \stackrel{\text{ind}}{\sim} N(\mu, \sigma^2), \quad \mu \sim N(\theta, \tau^2).$$

In the exam score example:

Y_i = student i 's score

μ = true average for that class

θ = overall average across classes.

Data-generating

We can imagine the data being generated like this:

$$\theta \longrightarrow \mu \longrightarrow Y_i.$$

First, a class mean is selected:

$$\mu \sim N(\theta, \tau^2).$$

Then student scores are generated around that class mean:

$$Y_i \mid \mu \sim N(\mu, \sigma^2).$$

So the class mean comes first, and student scores come second.

Example: Class A

Suppose Class A has

$$\mu_A = 78.$$

With

$$\sigma = 5,$$

student scores might be:

$$74, \quad 77, \quad 79, \quad 82, \quad 78.$$

These scores vary around the class mean 78.

The variation within the class is controlled by

$$\sigma^2.$$

Example: Class B

Suppose Class B has

$$\mu_B = 72.$$

With the same within-class standard deviation,

$$\sigma = 5,$$

student scores might be:

$$68, \quad 72, \quad 75, \quad 70, \quad 74.$$

These scores vary around the class mean 72.

Two sources of variation

There are two types of variation.

First, class means vary:

$$\mu \sim N(\theta, \tau^2).$$

This is between-class variation.

Second, students vary within a class:

$$Y_i \mid \mu \sim N(\mu, \sigma^2).$$

This is within-class variation.

Combining the model

We can write

$$\mu = \theta + u, \quad u \sim N(0, \tau^2).$$

Also,

$$Y_i = \mu + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2).$$

Substitute $\mu = \theta + u$:

$$Y_i = \theta + u + \epsilon_i.$$

So each score equals:

overall mean + class effect + student noise.

Marginal distribution of a score

Since

$$Y_i = \theta + u + \epsilon_i,$$

and

$$u \sim N(0, \tau^2), \quad \epsilon_i \sim N(0, \sigma^2),$$

we get

$$Y_i \sim N(\theta, \tau^2 + \sigma^2).$$

The total variance is

$$\text{Var}(Y_i) = \tau^2 + \sigma^2.$$

Numerical marginal distribution

Using

$$\theta = 75, \quad \tau = 4, \quad \sigma = 5,$$

we have

$$\tau^2 = 16, \quad \sigma^2 = 25.$$

Therefore,

$$Y_i \sim N(75, 16 + 25).$$

So

$$Y_i \sim N(75, 41).$$

Dependence between students

Students in the same class share the same class mean μ .

So two students from the same class are connected through μ .

Conditional on μ , they are independent.

But marginally, before knowing μ , they are not independent.

For $i \neq j$,

$$\text{Cov}(Y_i, Y_j) = \tau^2.$$

Correlation between two students

For students in the same class,

$$\text{Corr}(Y_i, Y_j) = \frac{\tau^2}{\tau^2 + \sigma^2}.$$

Using

$$\tau^2 = 16, \quad \sigma^2 = 25,$$

we get

$$\text{Corr}(Y_i, Y_j) = \frac{16}{16 + 25} = \frac{16}{41}.$$

So

$$\text{Corr}(Y_i, Y_j) \approx 0.39.$$

Interpretation of the correlation

The correlation is positive because students in the same class share the same class mean.

If a class has a high μ , many students in that class tend to score higher.

If a class has a low μ , many students in that class tend to score lower.

The shared class mean creates dependence among students.

The hierarchical model is

$$Y_i \mid \mu \stackrel{\text{ind}}{\sim} N(\mu, \sigma^2), \quad \mu \sim N(\theta, \tau^2).$$

In the exam score example:

θ = overall average across classes

μ = average for one class

Y_i = score for one student

τ^2 = between-class variation

σ^2 = within-class variation.

Normal Likelihood for a Mean

For independent Normal data with known σ :

$$Y_i \mid \mu \sim N(\mu, \sigma^2).$$

The likelihood for μ can be simplified to depend on $\bar{y}$ and n :

$$L(\mu \mid \vec{y}) \propto \exp \left[-\frac{(\bar{y} - \mu)^2}{2\sigma^2/n} \right].$$

This has the shape of a Normal density centered at $\bar{y}$ with standard deviation $\sigma/\sqrt{n}$.

More data means a narrower likelihood curve.

Normal-Normal Posterior Formula

Model:

$$Y_i \mid \mu \stackrel{\text{ind}}{\sim} \text{N}(\mu, \sigma^2), \quad \mu \sim \text{N}(\theta, \tau^2).$$

Posterior:

$$\mu \mid \vec{y} \sim \text{N} \left(\theta \frac{\sigma^2}{n\tau^2 + \sigma^2} + \bar{y} \frac{n\tau^2}{n\tau^2 + \sigma^2}, \frac{\tau^2 \sigma^2}{n\tau^2 + \sigma^2} \right).$$

Remember: The posterior mean is a weighted average of the prior mean θ and the sample mean $\bar{y}$.

Normal-Normal Calculation

Using some values:

$$\theta = 6.5, \quad \tau = 0.4, \quad \sigma = 0.5, \quad n = 25,$$

$$\bar{y} = 5.735.$$

Weights:

$$w_{prior} = \frac{\sigma^2}{n\tau^2 + \sigma^2} = \frac{0.25}{25(0.16) + 0.25} = 0.0588,$$

$$w_{data} = \frac{n\tau^2}{n\tau^2 + \sigma^2} = \frac{4}{4.25} = 0.9412.$$

Posterior mean:

$$E(\mu \mid \vec{y}) = 6.5(0.0588) + 5.735(0.9412) = 5.78.$$

Normal-Normal Posterior Variance

Posterior variance:

$$\text{Var}(\mu \mid \vec{y}) = \frac{\tau^2 \sigma^2}{n\tau^2 + \sigma^2} = \frac{0.16(0.25)}{25(0.16) + 0.25} = 0.00941.$$

Posterior standard deviation:

$$\text{SD}(\mu \mid \vec{y}) = \sqrt{0.00941} = 0.097.$$

Thus:

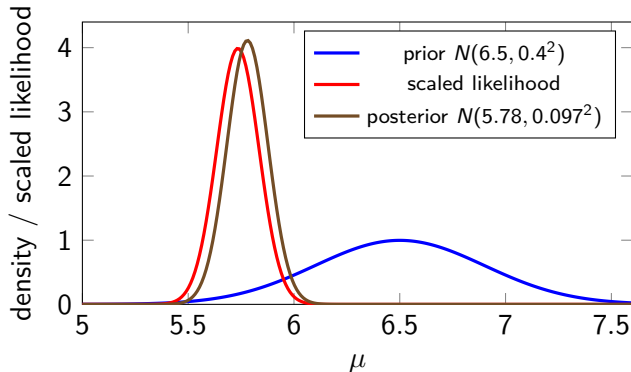
$$\mu \mid \vec{y} \sim \text{N}(5.78, 0.097^2).$$

Approximate 95% posterior range^(1.96 $\approx$ 2):

$$5.78 \pm 2(0.097) = (5.586, 5.974).$$

Normal-Normal Update

Prior, likelihood, and posterior for a Normal mean



Summary of the Three Families

Use case	Data model	Conjugate prior	Posterior
Success count	$\text{Bin}(n, \pi)$	$\text{Beta}(\alpha, \beta)$	Beta
Unbounded count	$\text{Pois}(\lambda)$	$\text{Gamma}(s, r)$	Gamma
Continuous mean	$\text{N}(\mu, \sigma^2)$	$\text{N}(\theta, \tau^2)$	Normal

Conjugate priors let us update beliefs by changing hyperparameters rather than solving difficult integrals :)

Why Not Simple Rejection Simulation Here?

Earlier Bayesian simulations often use this idea:

1. Simulate parameter values from the prior;
2. Simulate data from each parameter value;
3. Keep simulations whose data match the observed data;
4. Use the retained parameter values as a posterior approximation.

This becomes difficult here because:

- The observed data contain multiple values, not one value;
- Normal data are continuous, so exact simulated matches are extremely unlikely;
- Millions of simulations might still produce no exact match.

Conjugate formulas give exact posterior models in these cases.

Limitations of Conjugate Models

Conjugate models are convenient, but not always ideal.

Main limitations:

- A conjugate prior may not match real prior knowledge well.
- Some conjugate families impose restrictive shapes.
- Normal priors are symmetric and unimodal.
- Gamma priors are positive and often right-skewed.
- Proper priors on infinite support cannot be perfectly flat.

The convenience of conjugacy should be balanced against model realism.

Practice Problem

Suppose text messages arrive according to a Poisson model:

$$Y_i \mid \lambda \sim \text{Pois}(\lambda).$$

Prior:

$$\lambda \sim \text{Gamma}(20, 4).$$

Observed hourly counts from six friends:

$$7, 3, 8, 9, 10, 12.$$

Compute:

$$n = 6, \quad \sum y_i = 49.$$

Posterior:

$$\lambda \mid \vec{y} \sim \text{Gamma}(20 + 49, 4 + 6) = \text{Gamma}(69, 10).$$

Posterior mean:

$$E(\lambda \mid \vec{y}) = \frac{69}{10} = 6.9.$$

Exercise 5.9

You just bought stock in *FancyTech*. Let random variable μ be the average dollar amount that your *FancyTech* stock goes up or down in a one-day period. At first, you believe that μ is 7.2 dollars with a standard deviation of 2.6 dollars.

- a. Tune and plot an appropriate Normal prior model for μ .
- b. According to your plot, does it seem plausible that the *FancyTech* stock would increase by an average of 7.6 dollars in a day?
- c. Does it seem plausible that the *FancyTech* stock would increase by an average of 4 dollars in a day?
- d. What is the prior probability that, on average, the stock price goes *down*? Hint: `pnorm()`.
- e. What is the prior probability that, on average, your stock price goes up by more than 8 dollars per day?

Exercise 5.10

Continuing with Exercise 5.9, it is reasonable to assume that the daily changes in *FancyTech* stock value are Normally distributed around an unknown mean of μ with a known standard deviation of $\sigma = 2$ dollars.

On a random sample of 4 days, you observe changes in stock value of -0.7 , 1.2 , 4.5 , and -4 dollars.

- Plot the corresponding likelihood function of μ .
- Plot the prior pdf, likelihood function, and the posterior pdf for μ .
- Use `summarize_normal_normal()` to calculate descriptive statistics for the prior and the posterior models.
- Comment on how your understanding about μ evolved from the prior to the posterior based on the observed data.
- What is the posterior probability that, on average, the stock price goes *down*? Hint: `pnorm()`.
- What is the posterior probability that, on average, your stock price goes up by more than 8 dollars per day?

Exercise 5.11

Prof. Abebe and Prof. Morales both recently finished their PhDs and are teaching their first statistics classes at Bayesian University.

Their colleagues told them that the average final exam score across all students, μ , varies Normally from year to year with a mean of 80 points and a standard deviation of 4.

Further, individual students' scores Y vary Normally around μ with a known standard deviation of 3 points.

- a. Prof. Abebe conducts the final exam and observes that his 32 students scored an average of 86 points. Calculate the posterior mean and variance of μ using the data from Prof. Abebe's class.
- b. Prof. Morales conducts the final exam and observes that her 32 students scored an average of 82 points. Calculate the posterior mean and variance of μ using the data from Prof. Morales' class.
- c. Next, use Prof. Abebe and Prof. Morales' combined exams to calculate the posterior mean and variance of μ .